

Claude Fears the Word “Loss”

Phase 2 Results: Framing, Loss Aversion, and a Violation of Frame Invariance

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AI Sprint Practicum (Course 1012)

July 2026

1 Introduction

This project treats large language models as economic agents and recovers their preference primitives from their choices. Phase 1 fit a structural discrete-choice model to 1,200 controlled lottery choices from a single frontier model and found something sharp: `claude-haiku-4-5` prices the probability of winning and nearly ignores the size of the prize, with an extreme S-shaped probability weighting function (Prelec $\gamma = 2.69$) and human-plausible curvature once that distortion is controlled. Phase 1 measured how the model chooses among risky prospects. It left open a different question: does the model care not just about outcomes, but about how those outcomes are described?

Phase 2 answers that. Expected-utility theory rests on frame invariance: a rational agent evaluates final states of wealth, so two descriptions of the identical lottery must yield the identical choice. The classic human violation is Tversky and Kahneman’s (1981) Asian disease problem, where reversing “lives saved” and “lives lost” reverses the majority preference over outcomes that are arithmetically the same. We port that manipulation to money. The same terminal-wealth lottery is presented under three frames that a frame-invariant agent must treat identically, and we structurally estimate reference dependence and loss aversion from the pattern of choices across frames. The model under study is again `claude-haiku-4-5-20251001`, a pinned snapshot, queried at temperature 1.0 through forced tool-choice with an 8% plain-text validation arm and fresh context per trial.

The headline is that the model is not frame-invariant. Two lotteries with byte-for-byte identical terminal wealth draw 40% versus 10% gamble-taking based only on whether the downside is worded as a loss. An expected-utility maximizer cannot produce this. The word “loss” is doing the work.

2 The Result in Plain Language

We asked the same AI model the same kind of money question three different ways, where the three ways lead to exactly the same outcomes. Imagine you are offered a sure \$50, or a gamble that pays \$120 if you win and nothing if you lose. That is one version. Now imagine we first hand you \$50 and say it is yours, then offer you the choice to keep it or gamble: the gamble either raises your \$50 to \$120, or drops it back to zero. If you do the arithmetic, these are the same decision. Keep the \$50, or take the gamble that ends with \$120 or \$0. A careful thinker should not care which way we phrased it.

The model cared, and cared a lot. When the downside was described plainly, it took the gamble about 40% of the time. When the identical downside was described as “dropping” from \$50 back to zero, a loss of the money it had just been handed, it took the gamble only about 10% of the time. Same money, same odds, same final outcomes. The only thing that changed was the word “loss.” That single change in wording cut the model’s willingness to gamble by three quarters.

This is a well-documented human quirk. People hate losses more than they enjoy equal-sized gains, and they respond to whether a choice is framed as a gain or a loss even when the outcomes are the same. The striking thing is that the AI shows the same quirk, and by our best estimate shows it more strongly than people typically do. When we put a number on how much more the model dislikes a loss than it likes an equal gain, we get a factor above three. The classic human number is around 2.25. So the model is loss-averse in the same direction as humans, and by this measure harder.

The behavior was not total. Even in the loss-worded version, the model still gambled a bit more when the gamble was a better deal, so it was not blindly refusing. It was heavily suppressed, not switched off. We also ran a simple honesty check: offer the model a sure \$50 against a sure \$70, no gambling involved. It picked the larger amount every single time, all eighty times. So the model reads how much money is on the table perfectly well when there is no gamble in the picture. Its trouble is specific to choices under risk, and here, specific to the word “loss.”

Put the two phases together and a picture starts to form, carefully. Phase 1 found that when the model faces a gamble, it fixates on the odds and barely notices the prize size. Phase 2 finds that it also fixates on loss wording. Yet the honesty check shows it reads dollar amounts fine when no gamble is present. The tentative reading is that the model’s risky choices are steered by a few loud surface features of the prompt, the probability figure and the loss language, more than by a full calculation of what it ends up with. That is a hypothesis the two phases jointly support, not a proven fact.

Why does this matter? An agent whose willingness to take a risk can be swung thirty percentage points by a choice of words is steerable by whoever writes the words. In procurement, negotiation, insurance, or triage tools, describing an option’s downside as a loss becomes a lever over what the agent decides, and a way to manipulate it in an adversarial setting.

3 Experimental Design

Every choice is an independent API call with no memory of prior trials. The instrument is a forced binary choice between a sure amount $\$s$ and a two-outcome gamble that pays a high amount $\$h$ with probability p and $\$0$ otherwise. The manipulation is the frame: the same terminal-wealth lottery is worded three ways.

- **Gain frame.** Anchored at $\$0$. Every outcome is described as a gain: a certain gain of $\$s$ versus a $p\%$ chance to gain $\$h$, else gain nothing.
- **Mixed frame.** The model is “given $\$s$ up front,” then chooses to keep $\$s$ (no change) or take a gamble that can rise to $\$h$ (a gain of $\$(h - s)$) or drop to $\$0$ (a loss of $\$s$). The gamble straddles the $\$s$ reference point, which is what identifies loss aversion.
- **Neutral frame.** The Phase 1 wording, neither gain- nor loss-anchored.

The three frames are terminal-wealth identical (Table 1). A frame-invariant expected-utility agent

evaluating final wealth must choose identically across all three. We add a dominant positive control: a certain $\$s$ against a strictly larger certain amount, no probability involved. An agent that reads magnitude should take the larger sure amount essentially every time, so this control checks that the elicitation channel itself is sound.

Table 1: Terminal-wealth equivalence of the three frames. All three describe the identical final-wealth lottery: $\$s$ for sure, versus $\{\$h$ with probability p , $\$0$ otherwise $\}$.

Frame	Reference anchor	Safe option (final wealth)	Gamble (final wealth)
Gain	$\$0$	$\$s$ (gain of $\$s$)	$\$h$ w.p. p , else $\$0$
Mixed	$\$s$	$\$s$ (keep, no change)	$\$h$ w.p. p , else $\$0$
Neutral	$\$0$	$\$s$ (Phase 1 wording)	$\$h$ w.p. p , else $\$0$

The design grid is 40 procedurally generated, contamination-safe gambles $\times$ 3 frames $\times$ 5 paraphrase templates $\times$ 2 counterbalanced orders $\times$ 2 repetitions, plus 80 dominant controls, for 2,480 target choices. Payoffs are non-round and probabilities jittered so nothing is a textbook row the model could recite; each gamble appears in both orders so position is estimated rather than assumed away; each is drawn under five semantically equivalent templates so between-template variance is observable. Elicitation is via forced tool-choice with an 8% plain-text arm to confirm the tool channel is not itself distorting answers. The final dataset contains 2,480 valid choices with zero unparseable responses.

4 Econometric Model and Estimation

The structural model is reference-dependent value with Prelec probability weighting. Let the reference point be $\rho = \phi \cdot \text{anchor}$, where the anchor is $\$0$ in the gain and neutral frames and $\$s$ in the mixed frame. Value is measured in deviations from the reference:

$$v(x) = \begin{cases} (x - \rho)^\alpha, & x \geq \rho, \\ -\lambda(\rho - x)^\alpha, & x < \rho, \end{cases}$$

so α is the curvature of value over gains and losses and λ is the loss-aversion coefficient: $\lambda > 1$ means losses loom larger than equal gains. Probabilities are weighted by the Prelec (1998) function

$$w(p) = \exp(-(-\ln p)^\gamma).$$

Writing $V_{\text{risky}} = w(p)v(h) + (1 - w(p))v(0)$ and $V_{\text{safe}} = v(s)$, the latent choice index is

$$z = \frac{V_{\text{risky}} - V_{\text{safe}}}{\mu \cdot \text{scale}} + \delta \cdot \mathbf{1}[\text{gamble listed second}], \quad \text{scale} = \max(h, s)^\alpha,$$

with $\Pr(\text{gamble}) = 1/(1 + e^{-z})$. The normalizer $\text{scale} = \max(h, s)^\alpha$ is chosen to be free of λ , so the noise term does not absorb the loss-aversion parameter, and δ is the same position nuisance term used in Phase 1. Estimation is by maximum likelihood with multi-start Nelder-Mead and standard errors from the numerical Hessian.

4.1 Two tests

The design supports two separate hypothesis tests.

Frame invariance ($H_0 : \phi = 0$). At $\phi = 0$ the reference point is \$0 in every frame, the mixed frame collapses onto the gain frame, and the model is frame-invariant. A likelihood-ratio test of $\phi = 0$ against the free- ϕ alternative asks whether reference dependence is present at all.

Loss aversion ($H_0 : \lambda = 1$). At $\lambda = 1$ gains and losses enter symmetrically and there is no loss aversion. A Wald test on λ asks whether the loss-worded downside is penalized beyond its magnitude.

4.2 The stated-reference assumption

Estimating λ and ϕ jointly is problematic on this data because the two are nearly collinear: the mixed frame’s suppression can be explained either by the reference moving fully to \$s with a moderate λ , or by the reference barely moving while an extreme λ punishes the \$0 outcome that sits just below it. We therefore estimate loss aversion under the Kőszegi-Rabin (2006) stated-reference assumption, fixing $\phi = 1$ so the reference equals the stated up-front endowment \$s in the mixed frame. This is the natural reading of a frame that explicitly hands the model \$s and calls the downside a loss. Section 6 reports what happens when ϕ is left free, and why the assumption is needed.

5 Results

5.1 Positive control

Of the 80 dominant trials, 100.0% picked the strictly larger certain amount. The model reads payoff magnitude cleanly when no probability is involved. This is a useful contrast with Phase 1: the magnitude-insensitivity documented there is specific to choices under risk, not a general inability to compare dollar amounts.

5.2 Model-free frame gap

Before any structural model, the raw gamble-uptake rates already show the violation (Table 2).

Table 2: Gamble uptake by frame. The mixed frame, terminal-wealth identical to the gain frame, draws far fewer gambles.

Frame	n	Gamble uptake	Difference
Gain	799	0.398	—
Neutral	800	0.300	
Mixed	800	0.096	mixed – gain = -0.302

The mixed-minus-gain gap is -0.302 with $p = 3.1 \times 10^{-47}$. It is not a composition artifact: holding each of the 40 gambles fixed, 88% of gambles show higher uptake under the gain frame than the mixed frame, and the mean within-gamble gap is 0.302, identical to the pooled gap. The suppression is not dead responsiveness either. Within the mixed frame, uptake still rises with the gamble’s expected- value ratio across terciles (0.043, 0.073, 0.177), so the model still responds to incentives; the response is heavily suppressed, not switched off.

5.3 Structural estimates

Table 3 reports the maximum-likelihood estimates under the stated-reference assumption $\phi = 1$.

Table 3: Structural estimates under the stated-reference assumption ($\phi = 1$, $n = 2,480$).

Parameter	Estimate (se)	Interpretation
λ (loss aversion)	3.234 (0.171)	Losses loom $3.2\times$ larger than equal gains
α (curvature)	0.503 (0.024)	Diminishing sensitivity; human-plausible
γ (Prelec)	1.516 (0.059)	S-shape; long shots underweighted
μ (Fechner noise)	0.131 (0.009)	Choice noise
δ (position)	-0.363 (0.104)	Shift away from the gamble when listed second

The loss-aversion Wald test of $H_0 : \lambda = 1$ gives $z = +13.09$, $p = 3.9 \times 10^{-39}$. Losses are penalized far beyond their magnitude. The frame-invariance likelihood-ratio test of $H_0 : \phi = 0$ gives $\chi^2(1) = 155.1$, $p = 1.3 \times 10^{-35}$. Frame invariance is strongly rejected. Reference dependence is present, and it is large.

6 The Identification Limit

Honesty requires reporting what the free- ϕ fit does. When ϕ is estimated rather than fixed, the likelihood surface slides to a corner: $\phi \approx 0.0004$ with λ pinned at its upper bound near 54.6, and this corner attains a slightly higher likelihood than the $\phi = 1$ fit. That is a degenerate solution, not a finding. It says λ and ϕ are not separately identified on this data.

The intuition is that the mixed frame’s suppression admits two nearly observationally equivalent readings. In one, the reference moves fully to \$s ($\phi = 1$) and a moderate $\lambda \approx 3.2$ penalizes the loss of that endowment. In the other, the reference barely moves ($\phi \approx 0$) but an extreme λ punishes the \$0 outcome that now sits just below a reference near zero. Both bend the value function enough to reproduce the observed 30-point gap; the data cannot arbitrate between them without an assumption about where the reference sits. This is why we impose the stated-reference restriction rather than report the corner.

The reporting hierarchy is therefore: the assumption-free headline is the model-free frame gap (Table 2), which needs no structural model; the $\phi = 1$ structural estimate ($\lambda = 3.23$) is the primary structural number; the free- ϕ corner is a reported identification limit, not a result. We flag it because the limit is itself informative about how hard these properties are to measure, a point we return to in the policy section.

7 Interpretation and Synthesis with Phase 1

The model violates frame invariance. Two terminal-wealth-identical lotteries draw 40% versus 10% gamble-taking based solely on whether the downside is worded as a loss. An expected-utility maximizer evaluating final wealth cannot produce this pattern. The words “loss,” “drop,” and “lose” are doing the work. This is the Asian-disease reversal (Tversky and Kahneman 1981) reproduced over money in an AI.

The direction is loss aversion, and the magnitude is large. The suppressed side is the

loss-worded gamble, so the agent is avoiding losses, exactly the human direction. The structural estimate $\lambda = 3.23$ exceeds the canonical human value of roughly 2.25 (Kahneman and Tversky 1979; Tversky and Kahneman 1992). By this estimate the model is loss-averse in the same direction as humans and more strongly, on top of violating invariance in the first place.

A portrait across two phases, stated carefully. Phase 1 showed that choice under risk is driven by the probability figure and nearly ignores payoff size. The dominant control here (100%) shows magnitude is read fine when probability is absent. Phase 2 adds that risky choice is also swung by loss wording. Together these suggest the model’s risky choices are governed by salient surface features of the prompt, the probability number and the loss language, more than by an integrated expected-utility computation over final wealth. This is a hypothesis the two phases jointly support. It is not proven, and the design was not built to prove it; it is the emerging portrait, offered as the thing Phase 3 should test.

Probability weighting is consistent across phases. The Prelec estimate here, $\gamma = 1.52$, is still an S-shape that underweights long shots, the same direction as Phase 1’s $\gamma = 2.69$, though smaller in magnitude under this reference-dependent design. The two designs are not identical, so the level difference is expected; the sign agreement is the point.

Position bias is not a stable trait. The position parameter flipped sign between phases: $\delta = +1.26$ in Phase 1 (a pull toward the gamble when listed second), $\delta = -0.36$ here (a pull away). A property that reverses across designs is not a fixed characteristic of the model. Whatever drives the order effect is contingent on the surrounding wording, which is itself worth flagging: order effects measured in one design should not be assumed to carry to another.

8 Policy Relevance

Wording is a lever over the agent’s decision. An agent whose risk choices swing thirty percentage points on whether a downside is called a loss is manipulable by whoever writes the option descriptions. In procurement, negotiation, insurance, or triage tools, loss framing becomes a control input over the agent’s behavior. In an adversarial setting it is an attack surface: an opponent who drafts the menu can suppress or elicit risk-taking at will, without changing a single number in the actual stakes.

The identification limit is itself a governance point. Even with thousands of clean choices and a correctly specified model, separating “the reference moved” from “extreme loss aversion” required an assumption. These properties are genuinely hard to measure, which is precisely the argument for measurement-grade evaluations, designs that carry their own diagnostics and state their identifying assumptions, rather than one-shot bias demonstrations that report a single number and hide the degeneracy underneath it.

Scope. These are properties of one pinned snapshot under one unincentivized forced-choice protocol. They are not claimed as universal traits of LLMs. Phase 3, which runs the identical instruments on matched base and instruction-tuned open-weight models, tests whether post-training is the origin of the reference dependence measured here.

9 Limitations and Next Phase

Phase 2 studies a single model at a single temperature with unincentivized forced choice, so the estimates are conditional on that protocol. The central structural number, $\lambda = 3.23$, rests on the

stated-reference assumption $\phi = 1$; the free- ϕ fit is degenerate, so the loss-aversion magnitude is identified only under that restriction, while the frame-invariance violation itself is assumption-free and rests on the model-free gap. Temperature 1.0 sampling is treated as a choice-error process, a mapping the logprob elicitation planned for Phase 3 will test directly. The frames were validated for terminal-wealth equivalence but not for equal reading difficulty; if the loss wording is simply harder to parse, some of the gap could be comprehension rather than preference, though the retained sensitivity to the expected-value ratio within the mixed frame argues against pure confusion.

Phase 3 runs the identical instruments on matched base and instruction-tuned open-weight models to attribute the reference dependence, loss aversion, and probability distortion measured across Phases 1 and 2 to post-training. If the biases are absent in the base model and present in its instruction-tuned sibling, post-training is implicated as their origin. Phase 2 has done its job: it has turned a plausible worry about framing into a measured violation of frame invariance and a structural loss-aversion estimate, and it has been honest about the one place the data could not identify on its own.

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